

Linear Regression Model Updating

Adding covariates, adding observations, and deleting observations without starting over

$$\hat{\beta} = (X^T X)^{-1} X^T y$$



Learning objectives

Model updating turns algebraic structure into fast, interpretable regression calculations.

ROADMAP

- Update a least-squares fit after adding one covariate using residualization and a rank-one projection update.
- Relate the update to the Frisch–Waugh–Lovell theorem and the extra sum of squares for the added covariate.
- Update regression coefficients when one new subject is added using the Sherman–Morrison identity.
- Delete one subject without refitting the entire model and derive the deleted residual formula.
- Recognize leverage as the quantity controlling the sensitivity of add/delete updates.

Why update a regression model?

MOTIVATION

Scientific workflow

A new predictor becomes available, a sensitivity analysis removes a case, or new observations arrive sequentially.

Computational workflow

Refitting from scratch discards structure already computed in $X^T X$, its factorization, residuals, and the hat matrix.

Theoretical payoff

Update formulas expose what actually changes: one direction in the model space, one rank-one contribution, or one leverage-adjusted residual.

The recurring theme:

$\text{new quantity} = \text{old quantity} + \text{low-rank correction}$

Baseline full-rank model

We begin with $X \in \mathbb{R}^{n \times p}$ of full column rank.

NOTATION

Least-squares estimator

$$\hat{\beta} = A^{-1} X^T y, \quad A = X^T X$$

Hat and residual matrices

$$H = X A^{-1} X^T, \quad M = I - H$$

Residual vector / SSE

$$e = M y = y - X \hat{\beta}, \quad SSE = e^T e$$

Two matrix identities will drive the updates:

Block inverse / Schur complement

Sherman–Morrison rank-one inverse

01

Adding a covariate

How one new column changes coefficients, fitted values, and sums of squares.

Add one new covariate z

COVARIATE

$$X_+ = \begin{pmatrix} X & z \end{pmatrix}$$

We want to fit $y = X\beta + z\gamma + \varepsilon$

Key question

What part of z is genuinely new after accounting for the columns already in X ?

If $z \in \mathcal{C}(X)$, adding it does not enlarge the model space. Otherwise it contributes exactly one new direction.

Residualize the new covariate against X

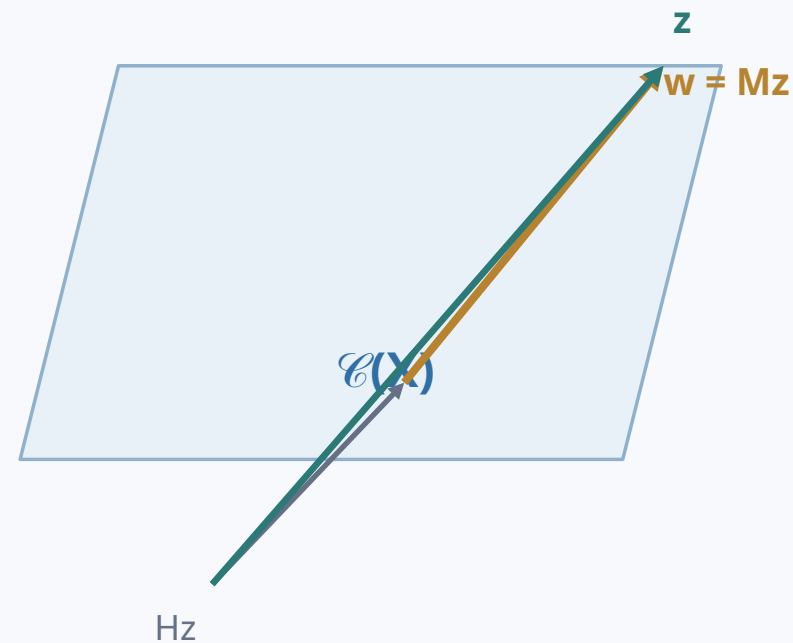
COVARIATE

$$w = Mz = (I - H)z$$

The vector w is the portion of z orthogonal to the existing model space.

Rank condition

The augmented design has rank $p+1$ exactly when $w \neq 0$, equivalently $z^T Mz > 0$.



Coefficient of the added covariate

COVARIATE

Once z is residualized against X , estimating y becomes a one-predictor regression:

$$\hat{\gamma} = \frac{z^T M y}{z^T M z} = \frac{w^T y}{w^T w} = \frac{w^T e}{w^T w}$$

Interpretation

Regress the old residuals e on the residualized new covariate w .

Nothing in z already explained by X can contribute to the new coefficient.

$$w \perp \mathcal{C}(X)$$

How the old coefficients change

COVARIATE

The added covariate can redistribute variation previously attributed to X .

$$\hat{\beta}_+ = \hat{\beta} - (X^T X)^{-1} X^T z \hat{\gamma}$$

Special case

If $X^T z = 0$, then the new covariate is already orthogonal to every old covariate, so $\hat{\beta}$ does not change.

The correction subtracts the portion of z that overlaps with the old design, scaled by the newly estimated effect $\hat{\gamma}$.

Frisch–Waugh–Lovell theorem

Partial regression is exactly the model update we just derived.

COVARIATE

- 1 Regress z on X** Keep residuals $w = Mz$.
- 2 Regress y on X** Keep residuals $e = My$.
- 3 Regress e on w** The slope is exactly $\hat{y}$ from the full model $[X \ z]$.

Why it matters

The coefficient of z measures its association with y after removing the linear information already contained in X .

Updating the projection matrix

COVARIATE

Because w is orthogonal to $\mathcal{C}(X)$, the augmented model space is an orthogonal direct sum:

$$\mathcal{C}(X_+) = \mathcal{C}(X) \oplus \text{span}(w)$$

Therefore the hat matrix receives one rank-one projection:

$$\mathbf{H}_+ = \mathbf{H} + \frac{ww^T}{w^T w}$$

Geometric meaning

Adding one linearly independent covariate adds exactly one new direction to the fitted-value space.

Updating SSE and the extra sum of squares

COVARIATE

$$M_+ = I - H_+ = M - \frac{ww^T}{w^Tw}$$

$$SSE_+ = SSE - \frac{(w^Ty)^2}{w^Tw} = SSE - \frac{(z^Te)^2}{z^TMz}$$

Extra sum of squares

The reduction in SSE is the squared projection of the old residual vector onto the new direction w .

$$SSR(z \mid X) = SSE - SSE_+$$

With one added covariate, the usual partial F statistic has 1 numerator degree of freedom and equals t^2 for testing $\gamma = 0$.

Block-matrix view of adding z

COVARIATE

$$X_+^T X_+ = \begin{pmatrix} X^T X & X^T z \\ z^T X & z^T z \end{pmatrix}$$

The Schur complement of $X^T X$ is

$$s = z^T z - z^T X (X^T X)^{-1} X^T z = z^T M z = w^T w.$$

Same quantity, three roles

s measures the independent variation in z , controls $\text{Var}(\hat{y} \mid X, z)$, and determines whether the augmented design has full rank.

$$\text{Var}(\hat{y} \mid X, z) = \frac{\sigma^2}{z^T M z}$$

Worked example: add a covariate

EXAMPLE

Reduced model: intercept + x, with

$$x = (0, 1, 2, 3)^T, \quad y = (1, 2, 2, 5)^T, \quad z = (0, 1, 0, 1)^T$$

Old fit

$$\hat{\beta} = \begin{pmatrix} 0.7 \\ 1.2 \end{pmatrix}, \quad SSE = 1.8$$

Residualized z

$$w = Mz = (-0.2, 0.6, -0.6, 0.2)^T$$

Update

$$\hat{\gamma} = 1, \quad \hat{\beta}_+ = \begin{pmatrix} 0.5 \\ 1.0 \end{pmatrix}$$

$$SSE_+ = 1.8 - \frac{0.8^2}{0.8} = 1.0$$

Extension: add several covariates at once

EXTENSION

$$X_+ = (X \quad Z), \quad W = MZ$$

$$\hat{\gamma} = (Z^T MZ)^{-1} Z^T M y = (W^T W)^{-1} W^T M y$$

$$H_+ = H + W(W^T W)^{-1} W^T$$

Conceptual rule

First remove from Z everything already explained by X. Then fit only the new directions.

The one-covariate formulas are the rank-one special case.

02

Adding a subject

One new row produces a rank-one update to $X^T X$.

Add one new subject

SUBJECT

Suppose a new observation arrives with covariate vector $x_\star \in \mathbb{R}^p$ and response $y_\star$.

$$X_+ = \begin{pmatrix} X \\ x_\star^T \end{pmatrix}, \quad y_+ = \begin{pmatrix} y \\ y_\star \end{pmatrix}$$

Then the cross-products update as

$$X_+^T X_+ = A + x_\star x_\star^T, \quad X_+^T y_+ = X^T y + x_\star y_\star.$$

Key structure

Only a rank-one matrix is added to $A = X^T X$.

Sherman–Morrison identity

SUBJECT

For invertible A and vectors u, v with $1 + v^T A^{-1} u \neq 0$,

$$(A + uv^T)^{-1} = A^{-1} - \frac{A^{-1}uv^T A^{-1}}{1 + v^T A^{-1}u}.$$

Set $u = v = x_\star$:

$$(A + x_\star x_\star^T)^{-1} = A^{-1} - \frac{A^{-1}x_\star x_\star^T A^{-1}}{1 + x_\star^T A^{-1}x_\star}$$

No full refit required

If A^{-1} or a factorization of A is already available, the new inverse is a low-rank correction.

Coefficient update after adding a subject

SUBJECT

Define the old-fit prediction error and prospective leverage:

$$r_{\star} = y_{\star} - x_{\star}^T \hat{\beta}, \quad h_{\star} = x_{\star}^T (X^T X)^{-1} x_{\star}.$$

Then

$$\hat{\beta}_{+} = \hat{\beta} + \frac{(X^T X)^{-1} x_{\star}}{1 + h_{\star}} r_{\star}$$

Two forces

A large residual pushes the coefficients more. A large $h_{\star}$ increases the denominator and reflects how strongly the new design point informs its own fit.

Reading the add-subject formula

SUBJECT

$r_\star$

Surprise

How far the new response is from the old model prediction.

$(X^T X)^{-1} x_\star$

Direction

Which combination of coefficients must move to accommodate the new subject.

$1/(1+h_\star)$

Damping

How the geometry of the existing design moderates the update.

Important

This $h_\star$ is computed from the old design. The new subject's leverage after inclusion is $h_\star/(1+h_\star)$.

SSE update after adding a subject

SUBJECT

The new residual sum of squares is

$$SSE_+ = SSE + \frac{r_\star^2}{1 + h_\star}$$

The residual of the new subject under the updated fit is

$$e_{\star,+} = \frac{r_\star}{1 + h_\star}.$$

Why not simply add $r_\star^2$?

The coefficients are allowed to move after the new subject is added. The model partially adapts to the new observation, reducing the increase from $r_\star^2$ to $r_\star^2/(1+h_\star)$.

Worked example: add a subject

EXAMPLE

Old simple regression:

$$\hat{\beta} = (0.7, 1.2)^T, \quad SSE = 1.8$$

Add $x_{\star} = (1, 4)^T$ with $y_{\star} = 6$.

Old-fit residual

$$r_{\star} = 6 - (0.7 + 1.2 \cdot 4) = 0.5$$

Prospective leverage

$$h_{\star} = x_{\star}^T A^{-1} x_{\star} = 1.5$$

Updated fit

$$\hat{\beta}_{+} = (0.6, 1.3)^T, \quad SSE_{+} = 1.9$$

$$1.8 + \frac{0.5^2}{1 + 1.5} = 1.9$$

03

Removing a subject

Leave-one-out calculations emerge from the same rank-one identity.

Delete observation i

DELETION

Let $\mathbf{x}_i^T$ be row i of $\mathbf{X}$, e_i its ordinary residual, and $h_i = h_{ii}$ its leverage.

$$\mathbf{A}_{(i)} = \mathbf{X}_{(i)}^T \mathbf{X}_{(i)} = \mathbf{A} - \mathbf{x}_i \mathbf{x}_i^T$$

Now the rank-one term is subtracted rather than added.

Condition

We need $1 - h_i > 0$. In ordinary full-rank regression with a nonessential observation, $h_i < 1$.

$$(\mathbf{A} - \mathbf{x}_i \mathbf{x}_i^T)^{-1} = \mathbf{A}^{-1} + \frac{\mathbf{A}^{-1} \mathbf{x}_i \mathbf{x}_i^T \mathbf{A}^{-1}}{1 - h_i}$$

Coefficient update after deleting subject i

DELETION

$$\hat{\beta}_{(i)} = \hat{\beta} - \frac{(X^T X)^{-1} x_i}{1 - h_i} e_i$$

Influence mechanism

A case has a large effect on $\hat{\beta}$ when it combines a large residual $|e_i|$ with high leverage h_i .

The denominator $1 - h_i$ can strongly magnify the effect of deleting a high-leverage observation.

$$\hat{\beta} - \hat{\beta}_{(i)} = \frac{(X^T X)^{-1} x_i}{1 - h_i} e_i$$

Deleted residual without refitting

DELETION

The leave-one-out prediction for observation i is $\mathbf{x}_i^T \hat{\boldsymbol{\beta}}_{(i)}$. Its deleted residual is

$$\mathbf{e}_{(i)} = \mathbf{y}_i - \mathbf{x}_i^T \hat{\boldsymbol{\beta}}_{(i)}.$$

Substituting the coefficient update gives the classic identity

$$\mathbf{e}_{(i)} = \frac{\mathbf{e}_i}{1 - h_i}$$

Key lesson

Ordinary residuals are too optimistic for predicting their own observations. Dividing by $1 - h_i$ removes that self-influence.

SSE deletion and PRESS

DELETION

$$SSE_{(i)} = SSE - \frac{e_i^2}{1 - h_i}$$

Deleting a case removes more than e_i^2 because the remaining coefficients are refitted.

The deleted-residual identity also gives leave-one-out prediction error immediately:

$$PRESS = \sum_{i=1}^n \left(\frac{e_i}{1 - h_i} \right)^2.$$

Computational payoff

All n leave-one-out predictions can be obtained from one fitted model.

Worked example: delete a subject

EXAMPLE

Return to the four-point regression with $\hat{\beta} = (0.7, 1.2)^\top$ and $SSE = 1.8$.

For observation 3: $\mathbf{x}_3 = (1, 2)^\top$, $e_3 = -1.1$, $h_3 = 0.3$.

Deleted coefficients

$$\hat{\beta}_{(3)} = \begin{pmatrix} 0.8571 \\ 1.3571 \end{pmatrix}$$

Deleted residual

$$e_{(3)} = \frac{-1.1}{0.7} = -1.5714$$

Deleted-data SSE

$$SSE_{(3)} = 1.8 - \frac{1.1^2}{0.7} = 0.0714$$

This subject is not high leverage, but its large residual makes it influential for the fitted line.

Connection to influence diagnostics

INFLUENCE

The delete-one coefficient change is the foundation of Cook's distance.

$$\begin{aligned} D_i &= \frac{(\hat{\beta}_{(i)} - \hat{\beta})^T X^T X (\hat{\beta}_{(i)} - \hat{\beta})}{p \text{ MSE}} \\ &= \frac{e_i^2}{p \text{ MSE}} \frac{h_i}{(1 - h_i)^2} . \end{aligned}$$

Two ingredients

Influence requires both discrepancy from the model (residual) and unusual position in predictor space (leverage).

Model updating turns that diagnostic interpretation into an exact algebraic statement.

04

Putting it together

One framework, three update problems.

Three updates, one pattern

SUMMARY

Add covariate	new column	$w = Mz$	$H + ww^T/(w^Tw)$
Add subject	new row	$r_\star, h_\star$	$\hat{\beta} + A^{-1}x_\star r_\star/(1+h_\star)$
Delete subject	remove row	e_i, h_i	$\hat{\beta} - A^{-1}x_i e_i/(1-h_i)$

Common structure

Every formula uses information from the current fit plus a low-rank correction driven by a residualized direction or a leverage-adjusted residual.

Computation: update factorizations, not explicit inverses

PRACTICE

- The formulas are written with $(X^T X)^{-1}$ because they are transparent theoretically; numerical software should usually avoid forming the inverse explicitly.
- QR and Cholesky factorizations admit stable row/column updates and downdates.
- Downdating is more delicate than updating: deleting a subject or covariate can expose near-singularity.
- If an added covariate has $z^T M z \approx 0$, it contributes almost no new direction and its coefficient variance becomes very large.
- Always track the rank of the updated design matrix.

What to be able to derive

THEORY

Adding z



Residualize $z \rightarrow$ derive $\hat{y} \rightarrow$ update $H \rightarrow$ update SSE.

Adding a subject



Update A and $X^T y \rightarrow$ apply Sherman–Morrison $\rightarrow$ simplify $\hat{\beta}_+$.

Deleting a subject



Use $A - x_i x_i^T \rightarrow$ derive $\hat{\beta}_{(i)} \rightarrow$ derive $e_{(i)} = e_i / (1 - h_i)$.

Exam-level emphasis

Understand where each update comes from and what it means geometrically; memorizing formulas without the projection/rank-one structure is not the goal.

Practice problems

PRACTICE

- Given X , y , and a candidate z , compute $w = Mz$ and obtain $\hat{y}$ without fitting the augmented regression directly.
- Show that $H_+ - H$ is idempotent of rank 1 when z adds a new direction.
- Starting from Sherman–Morrison, derive $\hat{\beta}_+$ for one newly added subject.
- Derive the deleted residual identity $e_{(i)} = e_i / (1 - h_i)$ from the coefficient deletion formula.
- Explain why an observation with a small residual can still matter if its leverage is high, and why high leverage alone does not guarantee a large Cook's distance.

Summary

TAKEAWAYS

Add a covariate

$$w = Mz, \quad \hat{\gamma} = \frac{w^T e}{w^T w}, \quad H_+ = H + \frac{ww^T}{w^T w}$$

Add a subject

$$\hat{\beta}_+ = \hat{\beta} + \frac{A^{-1}x_{\star}}{1+h_{\star}}r_{\star}, \quad SSE_+ = SSE + \frac{r_{\star}^2}{1+h_{\star}}$$

Delete a subject

$$\hat{\beta}_{(i)} = \hat{\beta} - \frac{A^{-1}x_i}{1-h_i}e_i, \quad e_{(i)} = \frac{e_i}{1-h_i}$$

The unifying idea is low-rank change.

Projection updates explain added covariates; rank-one inverse updates explain added and deleted observations.